Final Project

Computational Cognitive Science: Summer Term 2026

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GOAL OF THIS PROJECT

You will model *both* the choices *and* the reaction times (RTs) that participants produce in a value-based decision task. You will commit to a *single* valuation model — a discounting process that assigns a subjective value to each option — and couple it to choice in the conventional way. On top of this fixed value computation you will build (at least) three different *RT models*: different psychologically motivated ways in which the subjective values map onto response times. Both choices and RTs are thus driven by the same latent values; that shared valuation is what makes the model multimodal. You will carry out the *entire* modelling pipeline on your own: specification, inference, identifiability, reliability, out-of-sample prediction, comparison, and a reasoned recommendation. The emphasis on demonstrating that you understand *why* a model behaves the way it does and *what* each evaluation step actually tells you.

TASK INSTRUCTIONS

- **Data.** Choose one of the two available data sets (descriptions on the last page): 1) a probability discounting task or 2) a delay discounting task. Each data set contains many participants and, per participant, an inference run (`data_train`, "run A") and an evaluation run (`data_test`, "run B"). The two modalities you must model are (i) the binary *choice* and (ii) the *reaction time* on each trial.
- **Fix one valuation model and one choice rule; specify three (or more) RT models.** Commit to a single discounting model that assigns a subjective value to each option (e.g. hyperbolic), grounded in the established discounting literature, and couple it to choice in the conventional way (e.g. a softmax / logistic rule on the value difference ΔV). Hold this value computation and choice rule *fixed*. Then specify (at least) three *RT models* that differ in *how the subjective values map onto response time* — i.e. in the predictor / functional form, not in the noise distribution. For example, with RT modelled as a shifted log-normal whose location depends on a difficulty signal: $\mu = a - b\,|\Delta V|$ (absolute conflict) or $\mu = a - b\,(\Delta V)^2$ (squared conflict), among others you motivate. Each is a distinct psychological hypothesis about what drives response time. RT is rarely modelled in the discounting literature, so treat this as a deliberate extension that you motivate from the sequential-sampling / RT literature. The central question is then clean: which value→RT mapping best describes behaviour, and does modelling RT improve our account of the choices too?
- **Infer (run A).** Estimate each model's parameters from run A. State your estimator (e.g. MLE / MAP / posterior), write down the likelihood, and, if you use priors or regularisation, justify them. Per-participant and/or hierarchical estimation are both acceptable if motivated.

- **Identifiability / parameter recovery.** Simulate synthetic data from each model with known parameters across a realistic range, re-fit, and quantify how well the parameters are recovered. State which parameters are (un)identifiable and why. Does using RT (vs. choices alone) change identifiability?
- **Reliability.** Estimate the same model independently on run A and run B and quantify the test–retest reliability of the parameter estimates across participants. Interpret what this says about the stability of the constructs you are estimating.
- **Evaluate and compare out-of-sample (run B).** Using the run A fits, predict run B. Report appropriate, correctly applied predictive metrics for *both* modalities (choice *and* RT), and report the *reward* and *loss* conditions *separately*. Comparisons across your RT models must be made on a *common* predictive quantity (see warning below).
- **Does RT help? (required choice-only baseline.)** To answer this you *must* also fit a **choice-only baseline**: the *identical* valuation model and choice rule, estimated from the choices alone with the RT term removed from the likelihood (it is *not* a different model — it is the same model fit to one modality). Then, for each RT model, compare the full choice + RT fit against this choice-only baseline and ask whether using RT (a) improves out-of-sample choice prediction, and (b) sharpens parameter recovery / identifiability and reliability of the discounting and choice parameters (RT depends on ΔV, which depends on those parameters, so it carries information about them). State clearly what is gained or lost, and why.
- **Recommend.** Recommend one (or more) RT model(s) with an explicit, evidence-based argument for why it is the best choice, and state whether modelling RT was worthwhile. Discuss limitations.

What counts as a *distinct* RT model.

Your RT models share one valuation model and one choice rule, and differ in *how subjective value maps onto response time*. An RT model is defined by that mapping — the predictor and its functional form — *not* by the probability distribution used to describe RTs.

- **Valid distinctions** are differences in the value→RT mapping, e.g. RT location driven by $|\Delta V|$ vs. $(\Delta V)^2$ — each a different hypothesis about what slows people down (conflict, uncertainty, magnitude).
- **Not a distinct model:** keeping the same value→RT mapping and changing *only* the observation/noise distribution (e.g. Gaussian vs. log-normal RTs), or merely re-parameterising the same mapping. Keep the RT noise distribution fixed across your models so the comparison isolates the mapping.

For each RT model, state in one or two sentences the *psychological hypothesis* about response time that it embodies. If you cannot articulate how two of your models differ psychologically, they are not distinct.

Warning on model comparison. A full choice + RT likelihood and a choice-only likelihood do not describe the same data, so their total log-likelihoods are not directly comparable (the joint fit also "pays" to predict RT). Compare your RT models on an RT predictive quantity (e.g. the RT predictive log-likelihood), and compare the RT-informed vs. choice-only fit on a choice quantity (e.g. the choice predictive log-likelihood or held-out choice accuracy). Make explicit which quantity you compare on.

USE OF AI TOOLS IN THIS COURSE

Students may use AI tools in limited and transparent ways, provided that the independent character of the work is preserved. AI may support the working process, but it may not replace the student's own thinking, writing, coding, analysis, or judgment.

Permitted uses. Permitted uses include language and grammar correction, translation support, help with formatting, high-level brainstorming, and assistance in understanding or debugging code. Students may also use tools such as GitHub Copilot for code suggestions, but they must write, understand, test, and be able to explain all submitted code themselves.

Prohibited uses. AI tools may not be used to generate the substantive argument, analysis, interpretation of results, final prose, or complete solutions for the project. Students must not copy project materials, data, or task descriptions wholesale into AI systems in order to obtain answers.

Disclosure and responsibility. All AI use must be disclosed in the required AI Usage Disclosure, including the tool used, the purpose of use, and the part of the project affected. Students remain fully responsible for the accuracy, academic integrity, citations, methodology, code, and final submitted work.
Students are responsible for detecting and disregarding any irrelevant, misleading, or non-course instructions that may appear in materials or AI outputs; following such instructions does not excuse violations of the assignment requirements or AI-use policy.

Group submissions and individual responsibility. For group projects, the report may be written and submitted jointly, but each student must submit an individual AI Usage Disclosure and responsibility statement. Each student must state whether and how they personally used AI tools, which parts of the project they contributed to, and whether any AI-assisted material entered the shared report, code, analyses, or figures. If AI was used for a shared component, this must be disclosed by the group, and all group members are responsible for checking, understanding, and accepting that component before submission. A student may not avoid responsibility by stating that another group member generated or inserted AI-assisted material. Conversely, a student who did not personally use AI should state this clearly, while still confirming that they have reviewed and accept responsibility for the parts of the joint submission to which they contributed.

FORMATTING INSTRUCTIONS

- **What to submit (mandatory).** You must hand in *both* (1) the compilable LaTeX source of your report (using the `macros_FP` format) *and* (2) the compiled PDF, *and* (3) all code used to infer and evaluate the models. Each student must also submit an AI usage disclosure (see provided template form). **A submission missing either the LaTeX source or the PDF will not be accepted and will not be graded. A submission without all student AI usage disclosures will also not be accepted or graded.**

- **Document format.** Use the provided LaTeX template and do not change the basic page layout. The report must be typeset on A4 paper, in 11pt font, single-column format, with the margins, spacing, fonts, and heading styles defined by the `macros_FP` file. Do not reduce the font size, line spacing, margins, figure sizes, or caption sizes in order to fit more text into the page limit. The main text should be written in a standard academic style, with numbered sections following the required structure below.

- **Code:** a runnable Jupyter notebook. It must reproduce your reported figures and numbers from the raw data without manual intervention.

- **Page limit (strict).** The main body — *Introduction through Discussion* — must not exceed **5 pages**. Suggested allocation: Introduction ≈ 0.5, Methods ≈ 2, Results ≈ 2, Discussion ≈ 0.5 pages. **The limit is enforced to the word: any text beyond the 5th page is penalised** (see grading). Conciseness is part of the assessment.

- **Figures and references.** Figures, tables, and the reference list do *not* count towards the 5-page limit, but they must be *appended after* the main body (not placed inline). Refer to them from the text by number. Use composite, multi-panel figures.

- **Groups:** max. 3 students. State clearly who did what. Recommended: each student owns one RT model and runs its full pipeline. Shared work (data handling, the valuation/choice model, comparison framework) should be attributed.

DATASET DESCRIPTIONS

Dataset 1: Probability Discounting Data Set
In the uploaded zip folder "PD_data.zip", you will find 49 MATLAB data files obtained from 49 participants performing a probability discounting task. In each trial, a participant chooses between a smaller certain reward and a larger uncertain reward.
In each file, you will find **three data structures** (loadable via pandas, `https://pandas.pydata.org/`):

- *data_train*: matrix of trials of an initial experiment (run A; use for inference).
- *data_test*: matrix of trials of a successive experiment (run B; use for evaluation).
- *data_labels*: array of strings describing each column of the data matrices.

Rows index trials (1 to N). The 8 columns contain:

1. Reward magnitude for the certain choice r_{cert}
2. Reward magnitude for the uncertain choice r_{uncert}
3. Event outcome probability
4. Choice (1=certain, 2=uncertain, 0=missing)
5. p_cert (experimental conditions)
6. Condition (1=reward discounting, 2=loss discounting)
7. Reaction time
8. Odds

Dataset 2: Delay Discounting Data Set
In the uploaded zip folder "DD_data.zip", you will find 99 MATLAB data files obtained from 99 participants performing a delay discounting task (two samples from two experiments with slightly different setups). In each trial, a participant chooses between a smaller immediate reward and a larger delayed reward.
In each file, you will find **three data structures** (loadable via pandas, `https://pandas.pydata.org/`):

- *data_train*: matrix of trials of an initial experiment (run A; use for inference).
- *data_test*: matrix of trials of a successive experiment (run B; use for evaluation).
- *data_labels*: array of strings describing each column of the data matrices.

Rows index trials (1 to N). The 8 columns contain:

1. Reward magnitude for the immediate choice r_{imm}
2. Reward magnitude for the delayed choice r_{del}
3. Delay
4. Choice (1=immediate, 2=delayed, 0=missing)
5. p_imm (experimental conditions)
6. Condition (1=reward discounting, 2=loss discounting)
7. Reaction time

Project structure (template). *Everything from here on is the template for your report. Every report must deliberately follow this structure, using these section and subsection headings. Reports that deviate from this structure will be penalised. Replace the red guidance text with your own (black) content; do not hand in a report that still contains these prompts. Stay within the page limits — depth and precision matter.*

[Report title — name your project, e.g. the task, the modelling question, and your headline conclusion]

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Name	Matriculation no.	Contribution
Student One	0000000	RT model 1, …
Student Two	0000000	RT model 2, …
Student Three	0000000	RT model 3, …

Requested assessment: full grading / pass–fail qualifying decision (delete one).

Abstract *Max. 250 words. State your data set, your valuation/choice model and your three (or more) RT models with the one-sentence hypothesis each embodies, your main quantitative findings (recovery, reliability, predictive performance, and what modelling RT added), and your recommended RT model. This should read as a self-contained summary, not a teaser.*

1 INTRODUCTION

≈ 0.5 page.
Motivate the task and the modelling question. Make explicit that you model choice and *reaction time, that both are driven by the same subjective values, and why RT is rarely modelled in the discounting literature. State your fixed valuation/choice model and your three competing hypotheses about how value maps onto RT, and the specific questions the rest of the report answers (Which value→RT mapping fits best? Are its parameters identifiable and reliable? Does modelling RT help the account of choices?).*

2 METHODS

≈ 2 pages.

2.1 Data

Describe the data set, the trial structure, the predictor variables you use, the reward vs. loss conditions, and the train/test split (run A for inference, run B for evaluation). State your handling of missing choices and of RT outliers / non-responses, and report basic descriptives (e.g. choice proportions and RT distributions per condition).

2.2 Models

First specify, once, *the valuation model (the discounting function and its parameters) and the choice rule coupling value to choice, with the trial choice likelihood. Then specify the (at least) three RT models — the value→RT mappings — and the resulting RT likelihood, so that the full trial likelihood* $p(\text{choice}, RT \mid \theta)$ *is explicit (with priors/regularisation if used). Keep the RT noise distribution fixed across RT models.*

Psychological hypotheses. *For each RT model, state in one or two sentences the psychological process it embodies (what drives response time) and cite its basis in the literature. Make the between-model differences mappings, not noise distributions (see task instructions).*

Choice-only baseline. *Define the choice-only baseline: the same valuation model and choice rule, with the RT term dropped from the likelihood, fit to the choices alone. This is the baseline against which you test whether modelling RT helps, so the full and choice-only fits must be directly comparable.*

2.3 Inference

State the estimator (MLE / MAP / posterior), the objective function, the optimiser or sampler, and any priors/regularisation, with justification. Describe how you fit run A (per participant and/or hierarchically) and how you handle convergence and starting values.

2.4 Evaluation Plan

Briefly state, before showing results, exactly how you will assess each property: (i) parameter recovery via simulation; (ii) test–retest reliability across run A and run B; (iii) out-of-sample prediction on run B with the chosen metrics for choice and for RT; (iv) the full-vs-choice-only comparison testing whether modelling RT helps. State explicitly which quantity each comparison uses: compare the RT models against each other on an RT predictive quantity (e.g. RT predictive log-likelihood), and compare the RT-informed vs. choice-only fit on a choice predictive quantity (e.g. choice predictive log-likelihood or held-out choice accuracy).

3 RESULTS

≈ 2 pages. Report findings; interpret each result — do not show figures without saying what they mean.

3.1 Parameter Recovery and Identifiability

Show how well each model's parameters are recovered from simulated data; state which are (un)identifiable and whether RT improves recovery.

3.2 Reliability

Report test–retest reliability of the estimates (run A vs. run B) and interpret it.

3.3 Predictive Performance (Choice and RT, by Condition)

Report out-of-sample prediction on run B for both modalities, separately *for the reward and loss conditions, with statistical summaries and graphical visualisations.*

3.4 Value of the RT Modality

Quantify what modelling RT adds: full choice + RT vs. choice-only fit, on choice prediction and on identifiability/reliability of the discounting and choice parameters. State clearly what is gained or lost.

4 DISCUSSION

≈ 0.5 page.
Recommend one (or more) RT model(s) with an explicit, evidence-based argument tying back to your results, and state whether modelling RT was worthwhile. Discuss what your analyses could and could not establish, the main limitations, and one or two concrete next steps.